Program Paper

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Contents

1 Goal (Current Thought)

1. Show that prolongated associativity operad $\mathbb{P}\mathscr{M}$ is well-defined as a G-operad.

2. Promoe $\mathbb{P}\mathscr{M}$ promotes to a well-defined permutativity G-operad in Cat_G (say $\mathscr{E}\mathbb{P}\mathscr{M}$), and use it to define the new pseudoalgebras.

3. Show that $\mathscr{F}_{G,\mathbb{R}}$ pseudofunctors (if possible, G-functors, meaning preserving morphism sets' G equivariance) is well-defined, together with isomorphism to new pseudoalgebras as above.

4. (Not sure if it's true), but if we can show there is an equivalence of G-categories between the original $\mathscr{P}_G$ and $\mathscr{E}\mathbb{P}\mathscr{M}$, then maybe we cn also show the equivalence from the original categories.

The first three goals based on intuition and inspection seems to be true (However, my guess is this gives multiple restrictions on what the sets could be).

2 Introduction (ND)

3 Equivariance Preliminaries (Keep adding)

Given a group Π and element $\sigma \in \Pi$, we let $c_\sigma \in \mathrm{Aut}(\Pi)$ denotes the conjugation by σ, $c_\sigma(\psi) := \sigma\psi\sigma^{-1}$.

Definition 3.1. Fix a finite group G. Given Σ_n together with a group homomorphism $\alpha : G \to \Sigma_n$, the *semidirect product* $\Sigma_n \rtimes_{\alpha_c} G$ is the group with underlying set $\Sigma_n \times G$, together with product defined using G's conjugation action on Σ_n using α, as follow:

$$(\sigma, g) \cdot (\sigma', g') := (\sigma\ \alpha(g)\ \sigma'\ \alpha(g)^{-1}, gg')$$

Given a based finite G-set $\mathbf{n}^\alpha$, we ought to extend it to a semidirect product action:

Lemma 3.2. *The left G-action on $\mathbf{n}^\alpha$ extends to a $(\Sigma_n \rtimes_{\alpha_c} G)$-action, given by*

$$(\sigma, g)(i) := \sigma \circ \alpha(g)(i)$$

Proof. Fix $(\sigma, g), (\sigma', g') \in \Sigma_n \rtimes_{\alpha_c} G$, computation shows

$$(\sigma, g)[(\sigma', g')(i)] = (\sigma, g)[\sigma' \circ \alpha(g')(i)] = \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g')(i)$$

$$\begin{aligned}
[(\sigma, g) \cdot (\sigma', g')](i) &= (\sigma\ \alpha(g)\ \sigma'\ \alpha(g)^{-1}, gg')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g)^{-1} \circ \alpha(gg')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g)^{-1} \circ \alpha(g) \circ \alpha(g')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g')(i) \\
&= (\sigma, g)[(\sigma', g')(i)]
\end{aligned}$$

Also, it's clear identity acts trivially, verifying it's a group action. □

Another important notion is the twisted G-action if the space is given both G and Σ_n-action. For this, consider the following modification:

Definition 3.3. Given a group homomophism $\alpha : G \to \Sigma_n$, its *graph subgroup*

$$\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}$$

Using the first projection $\pi_1 : G \times \Sigma_n \twoheadrightarrow G$, it restricts to an isomorphism $\Gamma_\alpha \cong G$.

Given a $(G \times \Sigma_n)$-set X, define the *twisted G-action* on X by

$$g \cdot_\alpha x := (g, \alpha(g)) \cdot x$$

Which, the set with such twisted action is denoted as X^α.

Generalize the definition to arbitrary categories?

For further purpose, we consider the semidirect product action on each finite symmetric group:

Definition 3.4. Given Σ_n and a group homomorphism $\alpha : G \to \Sigma_n$, define a right $(\Sigma_n \rtimes_{\alpha_c} G)$-action on Σ_n

4 Categorical Notations (Keep adding, like statements on morphisms)

(Main references are Peter's papers).

Recall that $\mathscr{F}$ stands for the category of based finite sets, with objects being $\mathbf{n} := \{0, 1, ..., n\}$ (with 0 as basepoint), and homset $\mathscr{F}(\mathbf{n}, \mathbf{m})$ as all based maps between them. For generalization to equivariance setting for a fixed finite group G, one needs an extra G-equivariant structure.

Definition 4.1. Let $\mathscr{F}_G$ denotes the G-category of based finite G-sets, with objects being $\mathbf{n}^\alpha := \{0, 1, ..., n\}$ as set, together with a group homomorphism $\alpha : G \to \Sigma_n$; the homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) = \mathscr{F}(\mathbf{n}, \mathbf{m})$ as based set maps.

Here, we view $\Sigma_n \subseteq \mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{n}^{\alpha'})$ as permutation of the nonzero elements in $\mathbf{n}$. Which, the action on the homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is given by conjugation

$$g \cdot f := \beta(g) \circ f \circ \alpha(g)^{-1}$$

We'll use $G\mathscr{F} := [\mathscr{F}_G]^G$ as the G-fixed subcategory, which have the same objects as $\mathscr{F}_G$, while the homset $G\mathscr{F}(\mathbf{n}^\alpha, \mathbf{m}^\beta) = [\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)]^G$ are all the based G-maps between the two.

For the case $\mathscr{F}$, there are several subcategories, including Π (with all compositions of injections and projections), Λ (with all injections), $\Lambda^\perp$ (with all projections), Σ (with only bijections), and $\mathscr{E}$ (the effective maps). We make similar definition for $\mathscr{F}_G$:

Definition 4.2. Let $\Pi_G \subset \mathscr{F}_G$ be the subcategory where each morphism $\phi : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ has $|\phi^{-1}(j)| = 0$ or 1 for all $0 \neq j \in \mathbf{m}$.

Let $\Lambda_G \subset \Pi_G$ be the subcategory where each ϕ is injective (called *injections*); conversely, let $\Lambda_G^\perp \subset \Pi_G$ stands for the subcategory where each ϕ is surjective (called *projections*).

Let $\Sigma_G = \Lambda_G \cap \Lambda_G^\perp$, which is a subcategory with all ϕ being bijections.

Let $\mathscr{E}_G \subset \mathscr{F}_G$ be the subcategory where all morphism satisfies $\phi^{-1}(0) = \{0\}$, which these morphisms are called *effective*.

Similarly, we use $G\Pi, G\Lambda, G\Lambda^\perp$, and $G\mathscr{E}$ to denote the G-fixed subcategory for each (which homsets are limited to G-maps).

Insert some theorems (probably do some brief verification on them), keep increase them if needed.

However, there is an important notation we need:

Notation 4.3. Given a based finite G-set $\mathbf{n}^\alpha$, denote the group $\Sigma_{\mathbf{n}^\alpha} := G\Sigma(\mathbf{n}^\alpha, \mathbf{n}^\alpha)$, all the G-automorphisms of $\mathbf{n}^\alpha$.

5 Finitary G-Operads

5.1 Composability of Finite G-Sets

Here, each $\mathbf{j}^\alpha \in \mathscr{F}_G$ is a based finite set $\mathbf{j} = \{0, 1, ..., j\}$, together with a group homomorphism $\alpha : G \to \Sigma_n$.

Definition 5.1. The tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$ is called *composable*, if given any $g \in G$ and $r, s \in \{1, ..., k\}$, $\beta(g)(r) = s$ implies $\mathbf{j}_r^{\alpha_r} = \mathbf{j}_1^{\alpha_r}$.

In this case, denote $j := \sum_{r=1}^k j_r$, define the function $\gamma := \gamma(\beta; \vee_r \alpha_r) : G \to \Sigma_j$ by

$$\gamma(g)(i) := \alpha_s(g)(i)$$

where $\beta(g)(r) = s$ (permutation of blocks), $i \in \mathbf{j}_r$ (as the rth block in $\mathbf{j}$), and $\alpha_s(g)(i) \in \mathbf{j}_s$ (as the sth block in $\mathbf{j}$).

The composability is for composing the group homomorphisms into a larger group homomorphism:

Proposition 5.2. *If the tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$ is composable, then $\gamma(\beta; \vee_r \alpha_r)$ is a group homomorphism.*

Proof. Fix $r \in \{1, ..., k\}$, $g, h \in G$, let $s := \beta(g)(r)$, and $t = \beta(h)(s) = \beta(hg)(r)$. Now, the computation shows the following for $i \in \mathbf{j}_r^{\alpha_r}$, where r, i are arbitrary:

$$\gamma(g)(i) = \alpha_s(g)(i) \in \mathbf{j}_s$$

$$\gamma(h)(\alpha_s(g)(i)) = \alpha_t(h)(\alpha_s(g)(i)) = \alpha_t(h)(\alpha_t(g)(i)) = \alpha_t(hg)(i) \in \mathbf{j}_t$$

here we use the assumption where $\mathbf{j}_r^{\alpha_r} = \mathbf{j}_s^{\alpha_s} = \mathbf{j}_t^{\alpha_t}$. This indicates

$$\gamma(h) \circ \gamma(g)(i) = \gamma(h)(\alpha_s(g)(i)) = \alpha_t(hg)(i) = \gamma(hg)(i)$$

concluding $\gamma(h) \circ \gamma(g) = \gamma(hg)$. □

Here, we use $\epsilon_j : G \to \Sigma_j$ to denote the trivial group homomorphism. It satisfies unit axioms:

Proposition 5.3. *Given the composable tuples $(\mathbf{j}^\alpha; \mathbf{1}_1^{\epsilon_{1,1}}, ..., \mathbf{1}_j^{\epsilon_{1,j}})$ (with j copies of $\mathbf{1}$) and $(\mathbf{1}^{\epsilon_1}; \mathbf{j}^\alpha)$, then*

$$\gamma(\alpha; \vee_t \epsilon_{1,t}) = \alpha = \gamma(\epsilon_1; \alpha) : G \to \Sigma_j$$

Proof. For the first equality, for any $r \in \{1, ..., j\}$ and $g \in G$, let $s := \alpha(g)(r)$ (which, r, s can be viewed as the only element in the rth and sth block respectively, which we denote $1_r \in \mathbf{1}_r$, $1_s \in \mathbf{1}_r$ respectively). Then, the following equality holds:

$$\gamma(\alpha; \vee_t \epsilon_{1,t})(g)(1_r) = \epsilon_1(g)(1_s) = 1_s$$

Swapping back to normal labels, we get $\gamma(\alpha; \vee_t \epsilon_{1,t})(g)(r) = s$.

For the second equality, since there's only one block involved, G acts trivially on the block, hence $\gamma(\epsilon; \alpha)(g)(r) = \alpha(g)(r) = s$. The two equalities concluded $\gamma(\alpha; \vee_t \epsilon_{1,t})(g) = \alpha(g) = \gamma(\epsilon_1; \alpha)(g)$. □

Finally, it also satisfies associativity:

Proposition 5.4. *Given composable tuple* $(\mathbf{k}^{\delta}; \mathbf{j}_1^{\beta_1}, ..., \mathbf{j}_k^{\beta_k})$*, and each* $r \in \{1, ..., k\}$ *is associated with a composable tuple* $(\mathbf{j}_r^{\beta_r}; \mathbf{i}_{r,1}^{\alpha_{r,1}}, ..., \mathbf{i}_{r,j_r}^{\alpha_{r,j_r}})$ *(with* $\delta(g)(r) = t$ *implies the tuples are identical), then*

$$\gamma\left(\gamma(\delta; \vee_r \beta_r); \bigvee_r (\vee_s \alpha_{r,s})\right) = \gamma\left(\delta; \bigvee_r \gamma(\beta_r; \vee_s \alpha_{r,s})\right) : G \to \Sigma_i$$

Here, $\bigvee_r (\vee_s \alpha_{r,s})$ is wedging the $\mathbf{i}_{r,s}$ blocks in the order

$$\mathbf{i}_{1,1}, \mathbf{i}_{1,2}, ..., \mathbf{i}_{1,j_1}; \mathbf{i}_{2,1}, \mathbf{i}_{2,2}, ..., \mathbf{i}_{2,j_2}; ...; \mathbf{i}_{k,1}, \mathbf{i}_{k,2}, ..., \mathbf{i}_{k,j_k}$$

Also, observe that if $\delta(g)(r) = t$, then $\gamma(\beta_r; \vee_s \alpha_{r,s}) = \gamma(\beta_t; \vee_s \alpha_{t,s})$ (as we assume the tuples for r, t are identical), so the right map makes sense.

Proof. Denote the left map as γ_1, the right map as γ_2.

Given $\ell_{r,s} \in \mathbf{i}_{r,s}$ (r indicates the block for $\mathbf{j}$'s, and s indicates the block for $\mathbf{i}_r$'s), $g \in G$, suppose $\delta(g)(r) = t$ (implying $\mathbf{j}_r^{\beta_r} = \mathbf{j}_t^{\beta_t}$), and $\beta_r(g)(s) = \beta_t(g)(s) = w$, we get

$$\gamma(\delta; \vee_r \beta_r)(g)(r, s) = (t, \beta_t(g)(s)) = (t, w)$$

(Here, r indicates the rth block in $\mathbf{j}$, s indicates the sth element in $\mathbf{j}_r$).

Hence, γ_1 yields the following computation:

$$\gamma_1(g)(\ell_{r,s}) = \alpha_{t,w}(\ell_{t,w})$$

Similarly, if consider $\gamma(\beta_t; \vee_u \alpha_{t,u})(g)(\ell_{t,s}) = \alpha_{t,w}(g)(\ell_{t,w})$, hence

$$\gamma_2(g)(\ell_{r,s}) = \gamma(\beta_t; \vee_u \alpha_{t,u})(g)(\ell_{t,s}) = \alpha_{t,w}(\ell_{t,w})$$

due to the assumption $\delta(g)(r) = t$ for permuting the $\mathbf{j}$'s blocks. This indicates $\gamma_1(g) = \gamma_2(g)$ for all g. $\square$

5.2 G-Operad's Axioms

Instead of indexing over finite sets (objects in $\mathscr{F}$), the concept of G-operad is instead indexed over finige G-sets (objects in $\mathscr{F}_G$). Here, we fix $(\mathscr{V}, \otimes, \kappa)$ as our symmetric monoidal category.

Here the definitions are based on my personal interpretation, need verification!

Definition 5.5. A *G-operad* $\mathscr{C}_G$ in $\mathscr{V}$ consists of objects $\mathscr{C}_G(\mathbf{j}^\alpha)$ with a left G-action and a right Σ_j-action that combines to a right $(\Sigma_j \rtimes_{\alpha_c} G)$-action, unit morphism $\eta : \kappa \to \mathscr{C}_G(\mathbf{1}^{\epsilon_1})$, and structure morphism

$$\gamma_G : \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \to \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r \alpha_r)})$$

for each composable tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$. They must satisfy associative, unital, and equivariance in the following sense:

(a) Given composable tuple $(\mathbf{k}^\delta; \mathbf{j}_1^{\beta_1}, ..., \mathbf{j}_k^{\beta_k})$, and each $r \in \{1, ..., k\}$ provides another composable tuple $(\mathbf{j}_r^{\beta_r}; \mathbf{i}_{r,1}^{\alpha_{r,1}}, ..., \mathbf{i}_{r,j_r}^{\alpha_{r,j_r}})$ (with assumption $\delta(g)(r) = t$ implies the composable tuple for $\mathbf{j}_r^{\beta_r}$ and $\mathbf{j}_t^{\beta_t}$ are identical), let $j := \sum_r j_r$, $h_r := \sum_s i_{r,s}$, and $i := \sum_r h_r$, the following associativity diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\beta_r})\right] \otimes \left[\bigotimes_{s=1}^{j_r} \mathscr{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right] \arrow[r, "\gamma_G \otimes \mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\delta;\vee_r \beta_r)}) \otimes \left[\bigotimes_{r=1}^k \bigotimes_{s=1}^{j_r} \mathcal{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right] \arrow[d, "\gamma_G"] \\
 & \mathscr{C}_G(\mathbf{i}^{\gamma^2}) \\
\mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \left[\mathscr{C}_G(\mathbf{j}_r^{\beta_r}) \otimes \left(\bigotimes_{r=1}^{j_r} \mathscr{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right)\right]\right] \arrow[r, "\mathrm{id} \otimes (\gamma_G)^k"'] & \mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{h}_r^{\gamma(\beta_r;\vee_s \alpha_{r,s})})\right] \arrow[u, "\gamma_G"']
\end{tikzcd}$$

Here, $\gamma^2 := \gamma\left(\gamma(\delta; \vee_r \beta_r); \bigvee_r (\vee_s \alpha_{r,s})\right) = \gamma\left(\delta; \bigvee_r \gamma(\beta_r; \vee_s \alpha_{r,s})\right)$ based on **Proposition 3.4**.

(b) The following unit diagrams commute:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{j}^\alpha) \otimes \kappa^j \arrow[r, "\cong"] \arrow[d, "\mathrm{id} \otimes \eta^j"'] & \mathscr{C}_G(\mathbf{j}^\alpha) \\
\mathscr{C}_G(\mathbf{j}^\alpha) \otimes [\mathscr{C}_G(\mathbf{1}^{\epsilon_1})]^j \arrow[ur, "\gamma_G"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\kappa \otimes \mathscr{C}_G(\mathbf{j}^\alpha) \arrow[r, "\cong"] \arrow[d, "\eta \otimes \mathrm{id}"'] & \mathscr{C}_G(\mathbf{j}^\alpha) \\
\mathscr{C}_G(\mathbf{1}^{\epsilon_1}) \otimes \mathscr{C}_G(\mathbf{j}^\alpha) \arrow[ur, "\gamma_G"'] &
\end{tikzcd}$$

(c) Given composable tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$, the following $\Sigma_{\mathbf{k}^\beta}$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\mathrm{id} \otimes \sigma"] \arrow[d, "\sigma \otimes \mathrm{id}"'] & \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_{\sigma^{-1}(r)}^{\alpha_{\sigma^{-1}(r)}})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\gamma_G"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r \alpha_r)})
\end{tikzcd}$$

(d) Similar notation, the following $(\Sigma_k \rtimes_{\beta_c} G)$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^{\beta}) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "(\sigma,g)\otimes\mathrm{id}"] \arrow[d, "\gamma_G"'] & \mathscr{C}_G(\mathbf{k}^{\beta}) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G\big(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}\big) \arrow[r, "(\sigma(j_1,\dots,j_r),g)"'] & \mathscr{C}_G\big(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}\big)
\end{tikzcd}$$

(e) The following $\big((\Sigma_{j_1} \times \dots \times \Sigma_{j_k}) \rtimes_{(\alpha_1,\dots,\alpha_k)_c} G\big)$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^{\beta}) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\mathrm{id}\otimes((\tau_1,\dots,\tau_k),g)"] \arrow[d, "\gamma_G"'] & \mathscr{C}_G(\mathbf{k}^{\beta}) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G\big(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}\big) \arrow[r, "(\tau_1\oplus\dots\oplus\tau_k,g)"'] & \mathscr{C}_G\big(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}\big)
\end{tikzcd}$$

5.3 Prolongation of Associativity G-Operad, and Permutativity G-Operad

Definition 5.6. Define the *associativity G-operad* $\mathscr{M}_G$ in the category **GSet**, by $\mathscr{M}_G(\mathbf{n}^\alpha) := (\Sigma_n)^\alpha$, endows with the right action defined in **Equivariance Preliminaries**. Which, $\mathscr{M}_G = \mathbb{P}\mathscr{M}$ if consider prolongation.

Definition 5.7. Define the new *permutativity G-operad* $\mathscr{P}'_G$ in the category of small G-categories $\mathbf{Cat}_G$, as the chaotic prolongation $\mathscr{E}\mathbb{P}\mathscr{M}$ in $\mathbf{Cat}_G$.

Seems like Peter is trying to work on the original model now...

6 Finitely Generated G-Suboperads in $\mathscr{P}_G$

6.1 Suboperad $\mathscr{Q}_G$

This is based on the paper https://par.nsf.gov/servlets/purl/10298930, where they tried to give a finitely generated model for $\mathscr{P}_G$, and use an equivalent suboperad that is finitely generated.

Definition 6.1. Let T be a finite ordered G-set (so $T \cong \mathbf{n}^\alpha$ for $n = |T|$ and some $\alpha : G \to \Sigma_n$, if consider adding a basepoint). Denote the *external norm* $\otimes_T := \alpha : G \to \Sigma_n$.

For specific case $n = 0$, denote $e : G \to \Sigma_0$ (as unit) and for $n = 2$, denote $\otimes : G \to 1 \to \Sigma_2$ as the trivial group homomorphism into Σ_2.

For generation, I believe the paper tries to describe using transitive G-sets (as non-transitive finite sets, with multiple orbits, can be broken down into transitive ones; for this we need more precise formulation though).

Definition 6.2. A (based) finite G-set $\mathbf{n}^\alpha \in \mathscr{F}_G$ is called *transitive*, if it only has two orbits: $\{0\}$ and $\{1, ..., n\}$.

Definition 6.3. Let $\mathfrak{D}_G$ denotes a set of (based) ordered transitive G-sets, such that each isomorphism classes of transitive G-set contains exactly one copy. This is called *complete set of ordered G-orbits* in the paper.

More precisely, can view $\mathfrak{D}_G$ as a (chosen) subcategory of $\mathscr{F}_G$ (if considering not just equivariant maps).

As a side note, $\mathbf{D}_G$ is only unique up to isomorphism.

Some results I think we need (so I add in here):

Lemma 6.4. *For a finite group G, there are only finitely transitive finite G-sets, hence $\mathfrak{D}_G$ is finite.*

Proof. Let $n := |G|$ the order of the group. Consider any G-set $\mathbf{m}^\beta \in \mathscr{F}_G$ with $m > n + 1$. Then, it cannot be a transitive G-set: Assume it's transitive for the sake of contradiction, then for each $i \in \{2, ..., m\}$, there exists $g_i \in G$, such that $\beta(g_i)(1) = i$. By definition, each $i \neq j$ implies $\beta(g_i) \neq \beta(g_j)$ (as they map 1 to distinct elements), showing $g_i \neq g_j$. However, since $m > n + 1$, the collection $|\{g_2, ..., g_m\}| > n = |G|$, contradicting the cardinality of G. $\square$

This shows there are only finitely many group homomorphisms $G \to \Sigma_n$ that generates a transitive G-action on $\mathbf{n}$.

Another thing I wants to get familiar with is how to generate all finite G-sets (up to isomorphism) using these:

Lemma 6.5. *Up to some permutation, all (based) finite G-set $\mathbf{j}^\beta \cong \bigvee_{r=1}^k \mathbf{j}_r^{\alpha_r}$ in $\mathscr{F}_G$, with each $\mathbf{j}_r^{\alpha_r} \in \mathfrak{D}_G$.*

Proof. Let $O_1, ..., O_k$ be the (nonzero) G-orbits of $\mathbf{j}^\beta$, with cardinality $|O_r| = j_r$ (so, each $O_r \cong \mathbf{j}_r^{\alpha_r}$ for some transitive finite G-sets in $\mathfrak{D}_G$). Then, choose a permutation $\sigma \in \Sigma_j$ so that each O_r belongs to the rth ordered block in $\mathbf{j}$, this generates $\mathbf{j}^{c_\sigma \circ \beta}$, and $\mathbf{j}^{c_\sigma \circ \beta} = \bigvee_{r=1}^k \mathbf{j}_r^{\alpha_r}$ by this construction. $\square$

THis indicates to generate all such group homomorphisms, it suffices to consider the transitive ones, then the operadic structure will (possibly) help us glue everything together, and generate all the things we want.

Here is the operad we're considering:

Definition 6.6. In Set, let $Q_{G,\mathfrak{D}_G} := \langle \otimes_T \mid T \in \mathfrak{D}_G \rangle$, which is the G-suboperad in P_G generated by specific transitive group homomorphisms, operadic maps, and their translations using $(G \times \Sigma)$-actions.

In Cat_G, let $\mathscr{Q}_{G,\mathfrak{D}_G} := \mathscr{E} Q_{G,\mathfrak{D}_G}$ be the corresponding chaotic version of Q_G.

(Here, $Q_{G,\mathfrak{D}}$ is the notation in the paper).

6.2 Properties of $\mathscr{Q}_G$

Theorem 6.7. *For any finite group G and a complete set of ordered G-orbits $\mathbf{D}_{G,\mathfrak{D}_G}$, $\mathscr{Q}_{G,\mathfrak{D}_G}$ is an E_∞ G-operad, and the inclusion $\mathscr{Q}_{G,\mathfrak{D}_G} \hookrightarrow \mathscr{P}_G$ induces an equivalence of G-operads.*

Proof. cf. Bangs et. al. (the paper). $\square$

Theorem 6.8. *The above equivalens of G-operads induce a biequivalence between their respective 2-categories of algebras and lax maps.*

Proof. cf. Bangs et. al. (the paper again), unfortunately I'm trying to find the original source, $\square$

7 References